GEOAI 04 change report: Guwahati, Assam: regional check

Totals

Raw detections: 4932
Real changes after filters: 51
Area changed: 74.64 ha
Cropland affected: 4.62 ha (upper bound)
Built-up land affected: 1.89 ha
critical: 5
high: 5
medium: 9
low: 32

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Top 10 changes

#	Tier	Type	Area (ha)	Conf.	Reason
1	critical	water_loss	29.585	0.986	water receded: built-up index rose (+0.47), vegetation index rose (+0.19), water
2	critical	water_loss	17.97	0.986	water receded: built-up index rose (+0.54), vegetation index rose (+0.28), water
3	critical	water_loss	5.68	0.982	water receded: built-up index rose (+0.16), vegetation index rose (+0.26), water
4	critical	water_loss	3.76	0.982	water receded: built-up index rose (+0.28), vegetation index rose (+0.26), water
5	critical	built_up_gain	2.58	0.982	new construction / built-up: built-up index fell (-0.13), vegetation index rose
6	high	built_up_gain	2.15	0.984	new construction / built-up: built-up index fell (-0.06), water index rose (+0.0
7	high	water_loss	1.665	0.981	water receded: built-up index rose (+0.41), vegetation index rose (+0.22), water
8	high	water_loss	1.125	0.977	water receded: built-up index rose (+0.31), vegetation index rose (+0.24), water
9	high	water_loss	1.12	0.974	water receded: built-up index rose (+0.09), vegetation index rose (+0.32), water
10	high	built_up_gain	0.92	0.973	new construction / built-up: built-up index fell (-0.07), SWIR/blue ratio (paved

Sources and limits

- Copernicus Sentinel-2 L2A (ESA), via the Element84 Earth Search COG bucket
- ESA WorldCover 2021 v200 (10 m), CC-BY 4.0
- Optical imagery cannot see through cloud; flagged areas with cloud are removed, not analysed.
- Change type comes from spectral index rules (no training); check critical changes on the ground.
- 10 m pixels cannot resolve single small buildings or people.
- Rivers, sandbars and lake edges move between dates and can be flagged as change; bare soil in dry regions can look like construction. Check flagged areas near water.
- Land cover is ESA WorldCover 2021, newer than the before image (2019): land already cleared or under construction can be mislabelled as cropland, so treat cropland as an upper bound
- Buildings and road segments need OpenStreetMap, which was not reachable when this result was built: shown as unknown, not zero